

Enabling 24x7 Multi-Platform Infrastructure Support for a Global Financial Client

Introduction

A leading global financial services organization required an experienced technology partner to deliver round-the-clock L1/L2 infrastructure support for its hybrid server environment spanning Windows, Unix, and Linux systems.

With data centers distributed across the U.S., Europe, Asia, and Australia, and a growing footprint on Azure and AWS, the client's operations demanded continuous monitoring, proactive maintenance, and strict SLA adherence, including a 10-minute response time for P1 alerts.

Since 2016, NuWare has partnered with this client to provide comprehensive server administration, IAM operations, IT service desk support, and AWS platform management, ensuring maximum uptime, security compliance, and cost efficiency across the organization's mission-critical infrastructure.

The Challenge

The client's global infrastructure posed multiple operational and governance challenges that required an agile, scalable support framework:

1. **Hybrid Server Ecosystem**

A large mix of on-premise, virtualized, and cloud-based servers (Windows 2016/2019, Unix, and Linux) supporting databases, applications, and business systems.

2. **Global Datacenter Footprint**

Servers deployed across four continents, demanding synchronized support coverage and strict SLA compliance for incident response.

3. **Complexity of Patching and Maintenance**

Frequent security patching and OS upgrades required meticulous scheduling, testing, and rollback plans to minimize downtime.

4. **Continuous Monitoring and Alert Management**

Thousands of server alerts needed prioritization, acknowledgment, and escalation in near real-time.

5. **Identity and Access Management (IAM)**

Rapid onboarding and offboarding of global users necessitated precise control of security groups and access policies.

6. Cloud Operations and Cost Control

Managing cloud workloads on AWS and Azure, including patching, monitoring, scaling, and cost optimization, across multiple accounts.

NuWare needed to design a 24x7 multi-tiered support model capable of sustaining global operations, driving automation, and maintaining full compliance while handling a large ticket volume.

NuWare's Approach

NuWare deployed a dedicated offshore Infrastructure Support team of 8–10 engineers providing 24x7 global coverage across physical, virtual, and cloud environments. The support model integrated L1 monitoring, L2 remediation, IAM operations, IT service desk, and cloud infrastructure management, all governed through ITIL-aligned processes.

1. Server Administration (SA) – L1/L2 Support

Scope of Service:

Supporting over 897+ servers, including physical, virtual, and cloud instances, across technologies such as Windows Server, VMware, Citrix, AD, Azure, AWS, and Linux.

Core Activities:

- Round-the-clock server monitoring with SLA-based acknowledgment for P1 alerts.
- Patch Management & Firmware Upgrades for Windows, Unix, and Linux environments.
- Disaster Recovery (DR) & Failover Testing using custom scripts and automation workflows.
- Health Checks & Maintenance: Regular CPU, memory, and storage analysis.
- Scheduled Change Requests: Controlled rollout of system updates and configuration changes.
- Batch Job & Process Monitoring: Identifying slowness or failures and escalating accordingly.
- Certificate Renewals & Server Reboots based on predefined uptime thresholds.
- Security Monitoring: Managing Defender security portals, updating virus definitions, and ensuring endpoint compliance.
- Cloud Resource Analysis: Monthly cost reviews for AWS and Azure workloads.
- Documentation: Creating and updating SOPs, runbooks, and knowledge base entries.

- **Proactive Capacity Planning:** Analyzing trends to right-size hardware and optimize cloud costs.
- **Alert Notification Configuration:** Automating server and application alerts via Netcool, Splunk, and vRealize.
- **Reporting:**
 - Server utilization and compliance reports.
 - Website and application downtime analysis.
 - Monthly cloud cost and resource utilization dashboards.

Average Ticket Volume: ~200 tickets per month handled by the SA team.

2. IAM Operations

NuWare's IAM support team managed all user lifecycle activities, from onboarding to access termination, ensuring secure and auditable identity governance.

Key Activities:

- **User Onboarding:** Creating new user IDs with appropriate geo-based profiles and security group mapping.
- **Group Management:** Adding/removing users from security groups after manager approvals, creating new groups, and flushing memberships as needed.
- **User Termination:** Timely revocation of access across applications, networks, and storage systems.
- **Ticket Handling:** ~200 IAM-related tickets monthly.

All IAM processes followed least privilege principles, integrated with Active Directory (AD) and Azure Active Directory (AAD) for unified identity management.

3. IT Service Desk / Service Delivery Support

NuWare provided 24x7 IT service desk coverage, handling over 5,000 tickets per month, ensuring smooth coordination across multiple technical teams.

Scope of Service:

- **Password/MFA Reset:** Secure resets post user verification.
- **Troubleshooting & Redirection:** Routing issues to respective L2 teams — End User Support, Workspace Ops, Storage, Network, Security, or Application teams.
- **Incident & Problem Management:**
 - Creating and tracking incident (INC) and problem (PRB) tickets.

- Coordinating resolution and sending **P1/P2/P3 notifications**.
- Generating and analyzing monthly trend reports to identify recurring issues.
- **Service Delivery Reporting:** Monitoring SLA adherence and identifying service bottlenecks.

4. AWS Cloud Infrastructure Support

NuWare's cloud team provided 8x7 AWS support for the client's U.S.-based nutrition and wellness division, managing a direct-to-consumer e-commerce environment.

AWS Services Supported:

- **VPC, ECS, ECR:** Dockerized microservices for scalable container management.
- **ALB (Application Load Balancer):** Auto-scaling based on internet traffic spikes.
- **RDS, S3, ElastiCache, OpenSearch:** Data storage, caching, and indexing services.
- **Monitoring & Alerting:**
 - **CloudWatch** and **New Relic** for real-time performance visibility.
 - **PagerDuty** for automated escalation within 10 minutes if alerts go unacknowledged.
- **Collaboration with Dev & DBA Teams:** Supporting code deployments, release management, and stage environment validation.
- **Reporting:** Daily monitoring reports generated from CloudWatch and New Relic.

Average Monthly Volume: 10–15 change control (CC) tickets, with quick turnaround on high-priority issues.

Outcomes

The collaboration between NuWare and the client has delivered measurable and lasting results across operational efficiency, compliance, and performance.

1. High Availability and Performance

- Achieved **99.9% infrastructure uptime** across all datacenters and cloud workloads.
- Reduced downtime through proactive issue detection and automated remediation.

2. SLA Excellence

- Consistently maintained **<10-minute response time** for P1 alerts.

- Improved resolution time for high-priority incidents by 30%.

3. Operational Efficiency

- Managed over **5,000+ tickets/month** with streamlined processes and automation.
- Reduced manual monitoring efforts through script-based automation and alert optimization.

4. Audit and Compliance

- Delivered monthly **resource utilization, compliance, and downtime reports** for IT governance.
- Supported all **PWC and internal IT audits** with standardized documentation.

5. Cost Optimization

- Achieved **15–20% savings** through server rightsizing and proactive cloud cost analysis.

Future Outlook

NuWare is extending its partnership with the client to integrate next-generation infrastructure capabilities, including:

- **AI-Powered Monitoring:** Predictive analytics for performance and capacity forecasting.
- **Self-Healing Automation:** Automated remediation for failed processes or patch cycles.
- **Enhanced Observability:** Unified dashboards integrating Splunk, CloudWatch, and Netcool.
- **Cloud-Native Expansion:** Migration of additional workloads to Azure and AWS with full automation.

These initiatives aim to evolve the client's infrastructure from reactive operations to a self-optimizing, intelligent platform that delivers greater agility and resilience.